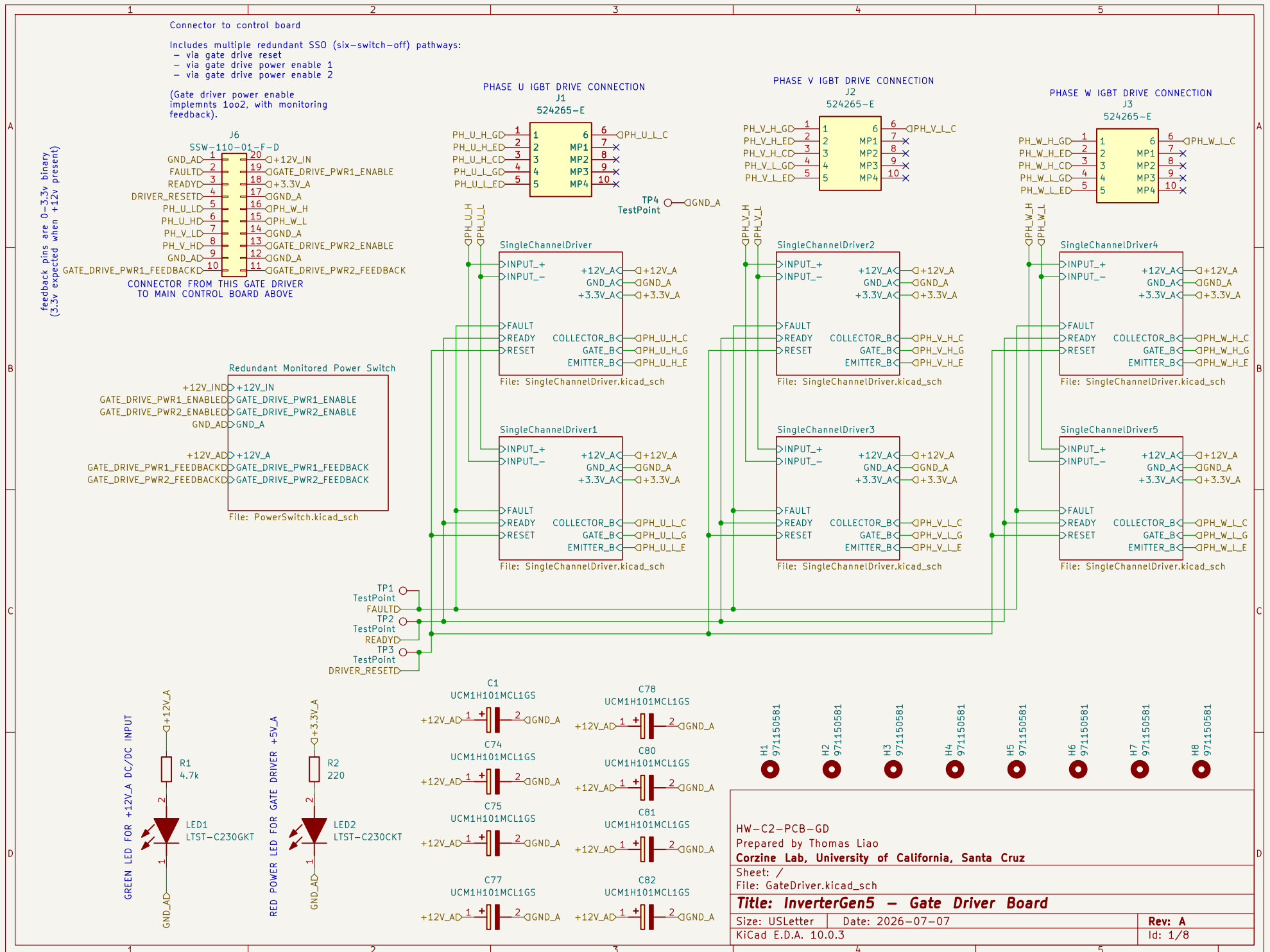
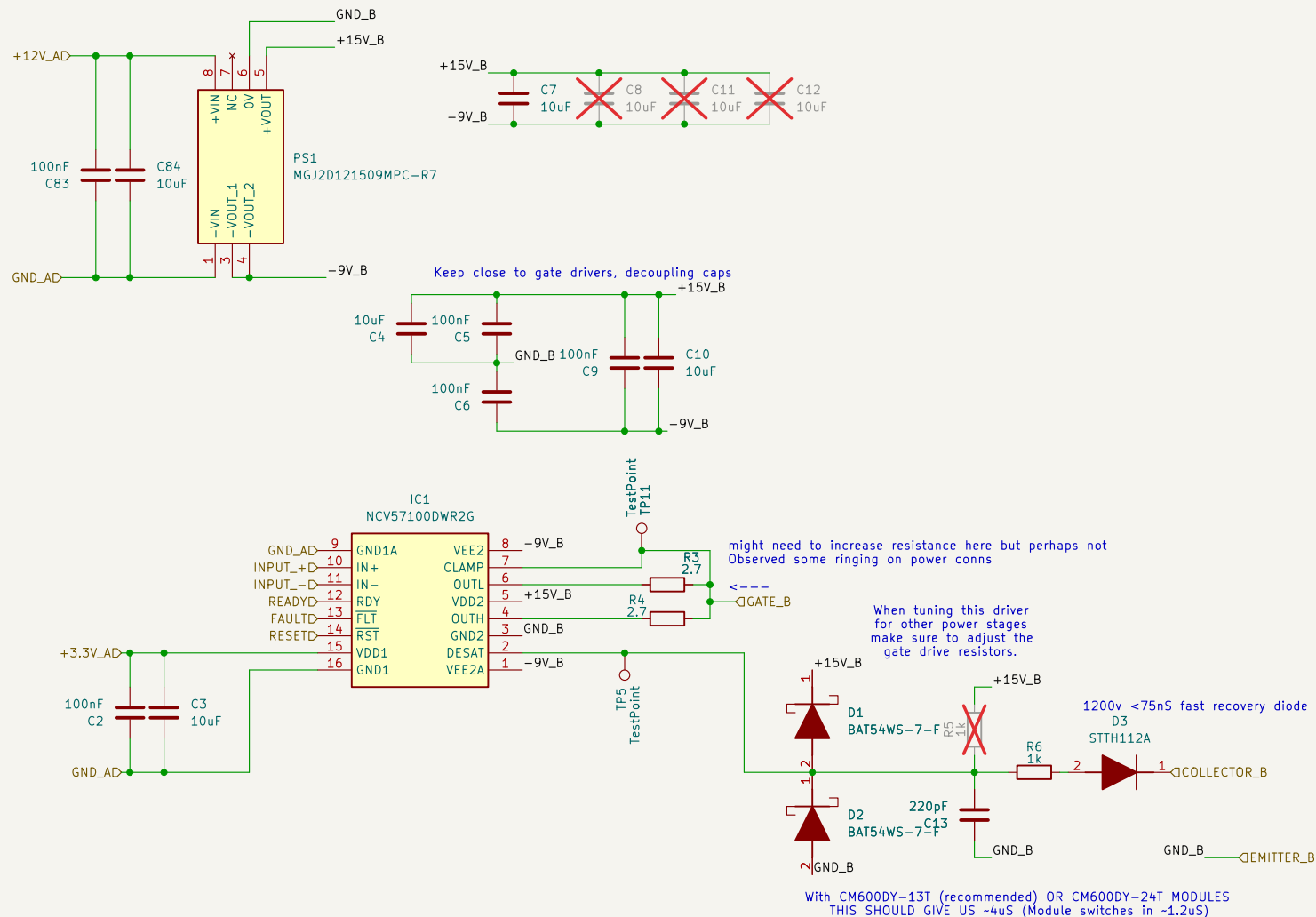






HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver1/

File: SingleChannelDriver.kicad_sch

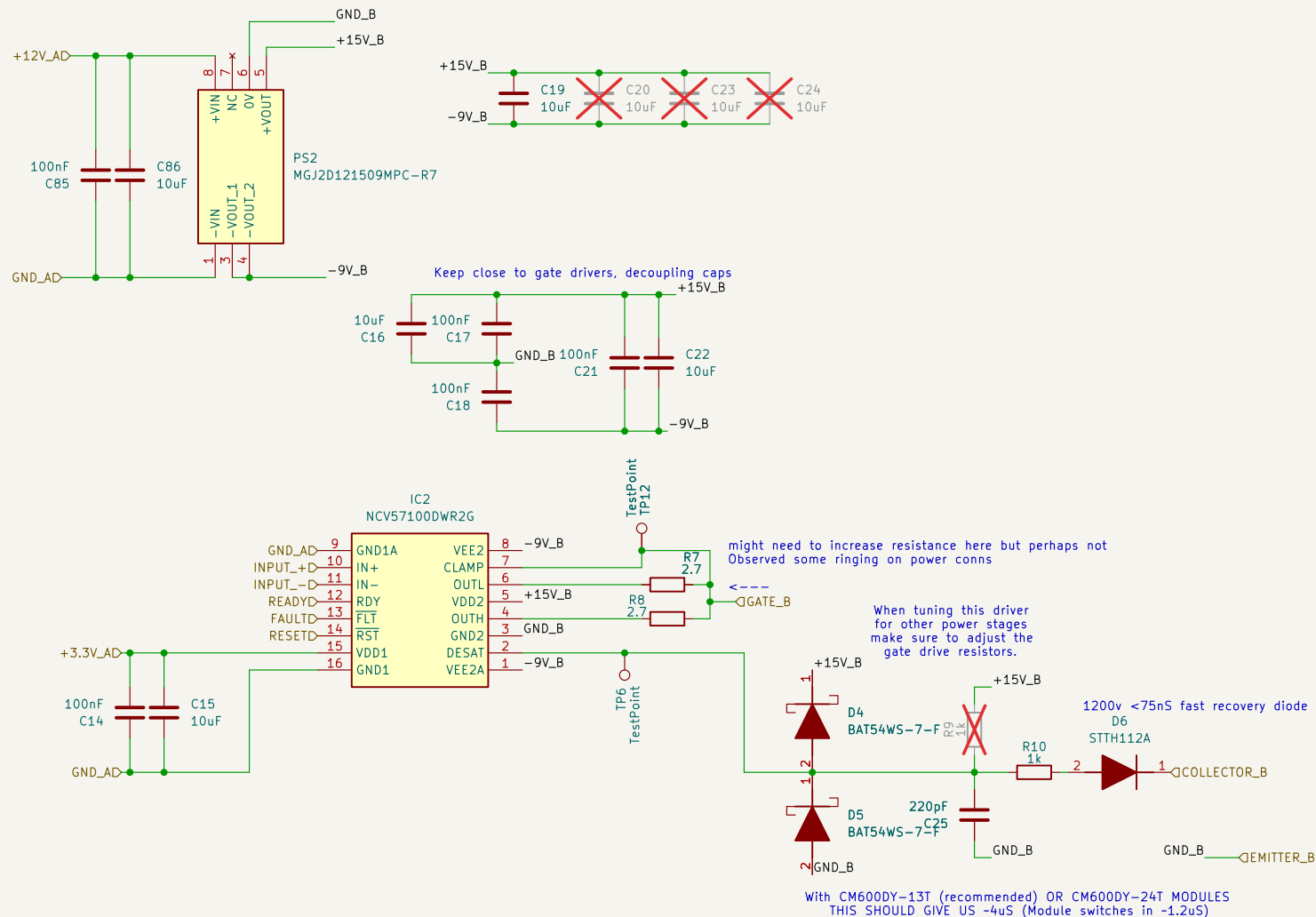
Title: InverterGen5 - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 2/8



HW-C2-PCB-GD

Prepared by Thomas Liao

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Sheet: /SingleChannelDriver2/

File: SingleChannelDriver.kicad_sch

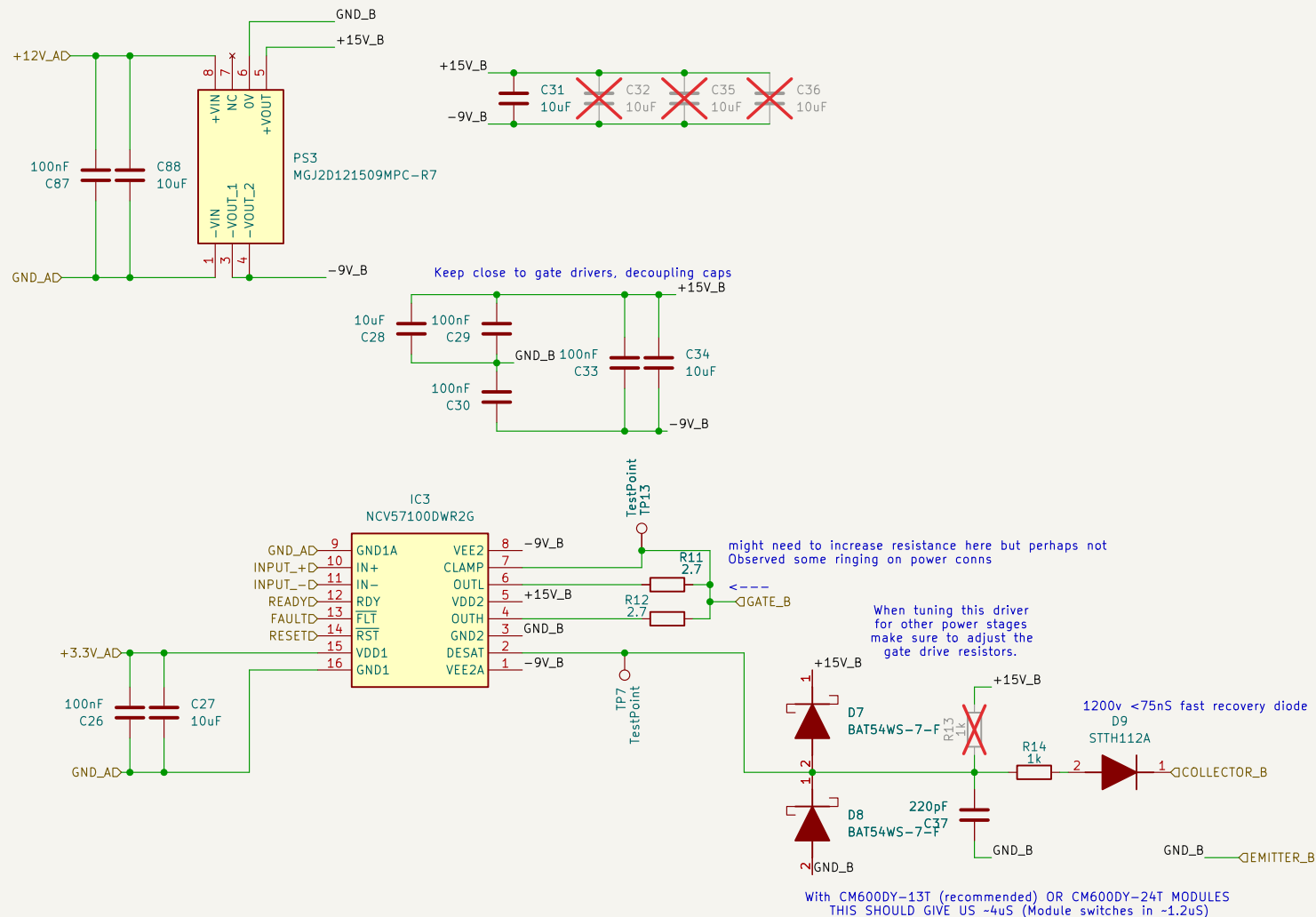
Title: InverterGen5 - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 3/8



HW-C2-PCB-GD

Prepared by Thomas Liao

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Sheet: /SingleChannelDriver/

File: SingleChannelDriver.kicad_sch

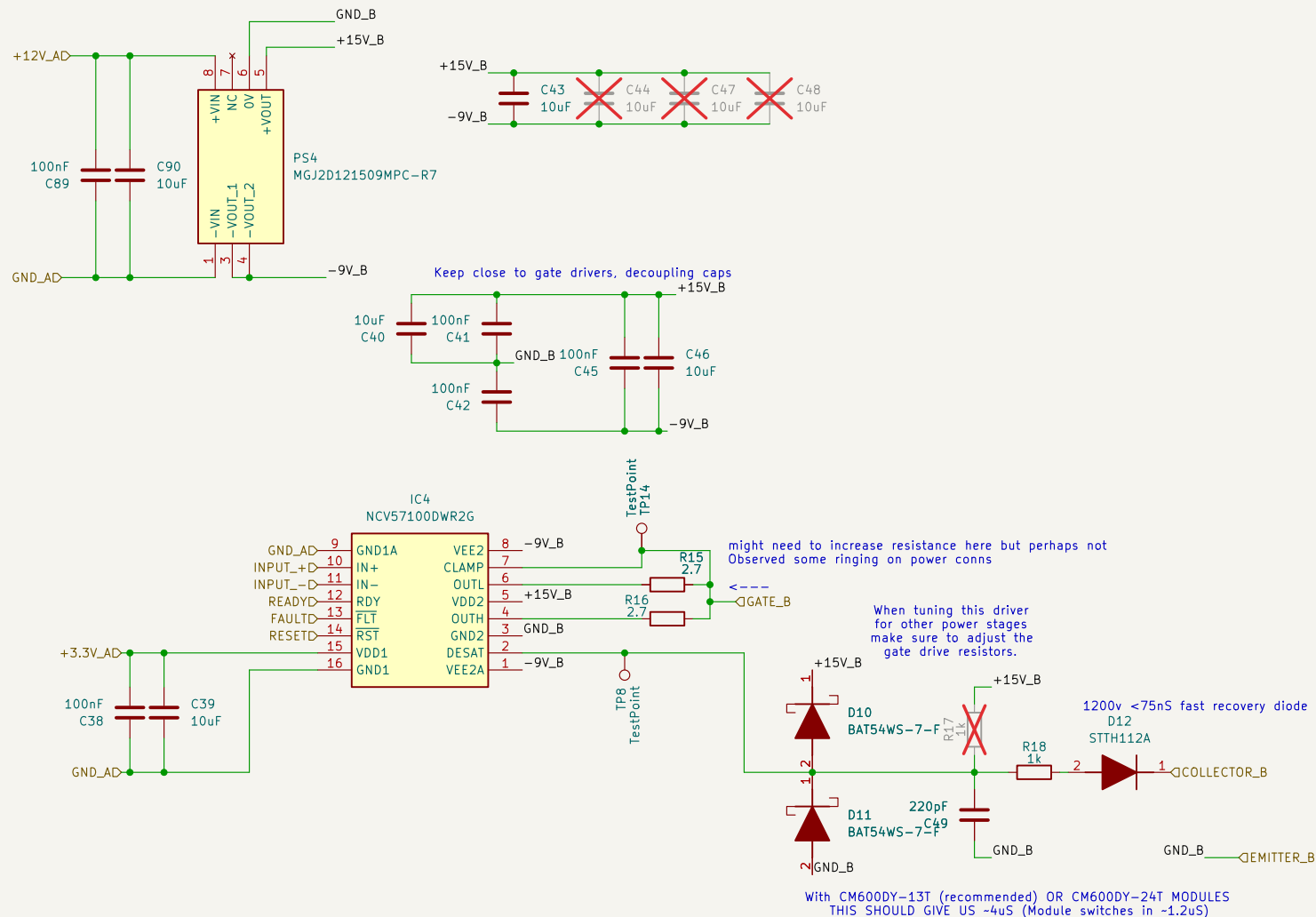
Title: InverterGen5 - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 4/8



HW-C2-PCB-GD

Prepared by Thomas Liao

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Sheet: /SingleChannelDriver3/

File: SingleChannelDriver.kicad_sch

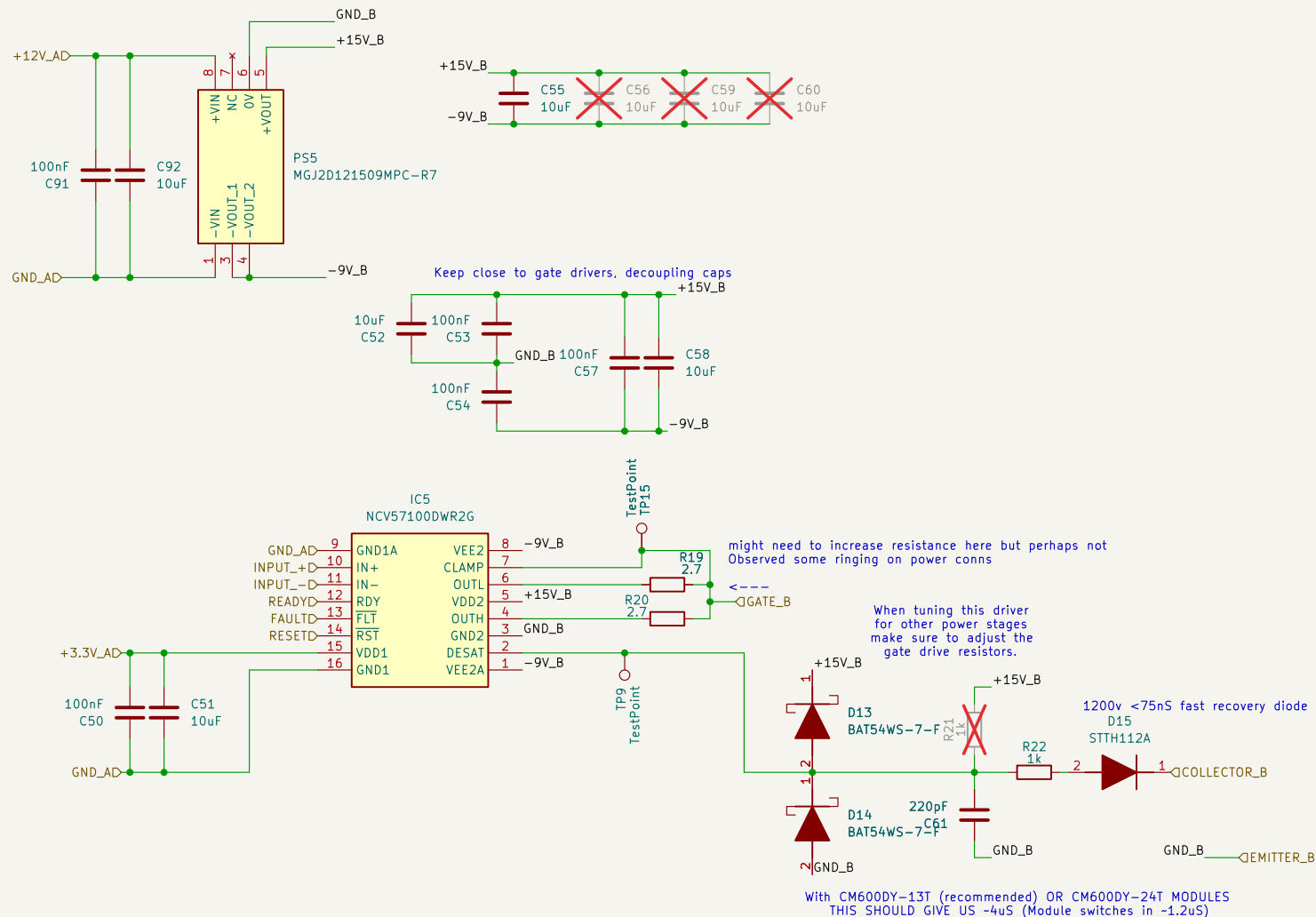
Title: InverterGen5 - Gate Driver Board

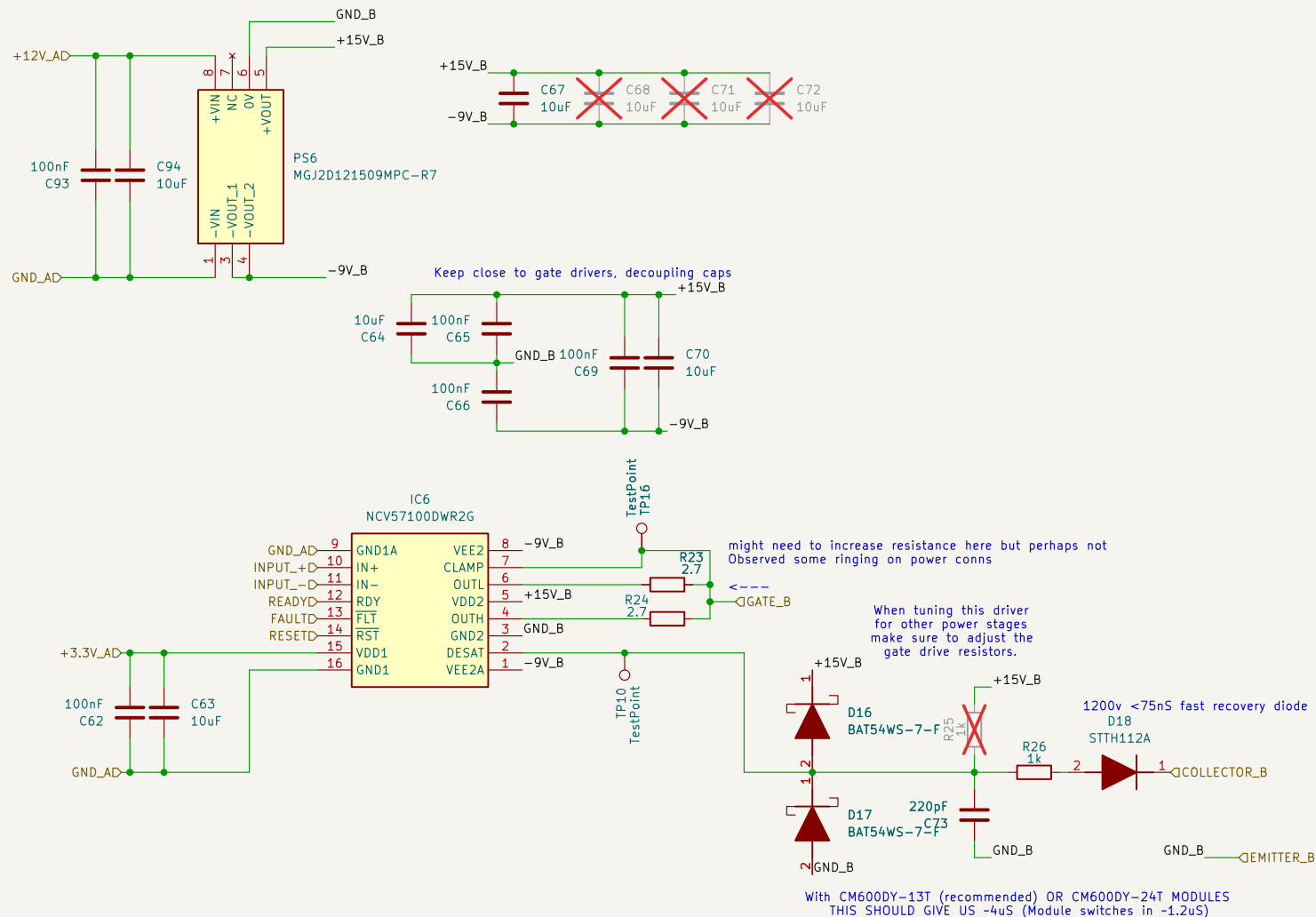
Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 5/8





HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver5/

File: SingleChannelDriver.kicad_sch

Title: InverterGen5 - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

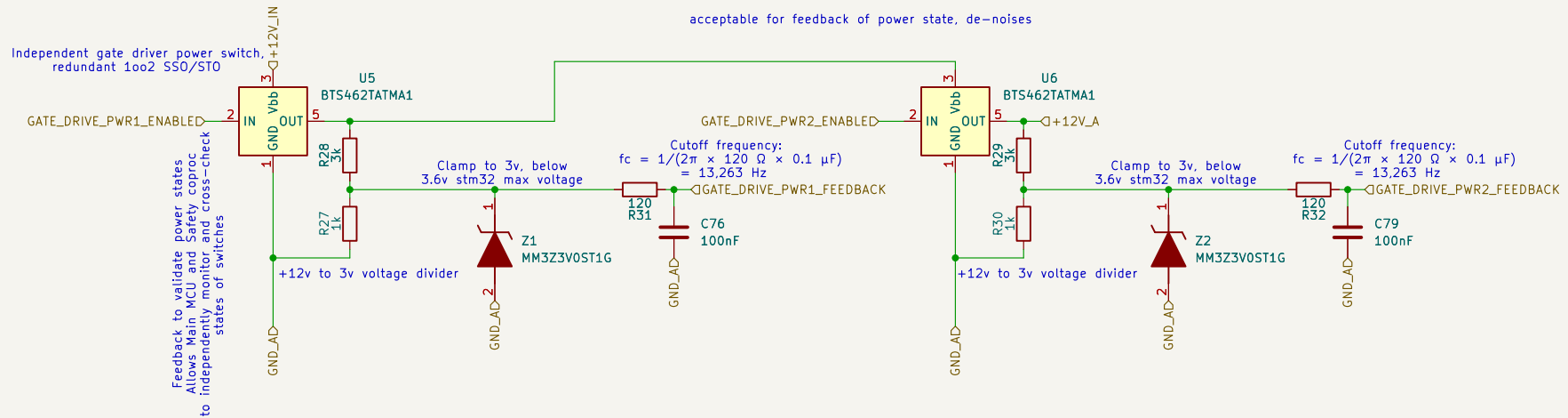
Id: 7/8

This circuit implements a redundant pathway for both mcus to implement Six Switch Open / Safe Torque Off in addition to regular pwm break and gate driver reset.

The following switches are independent systems with feedback monitoring. Each MCU cross monitors returned inputs, and can validate if the switch failed to respond as commanded – thus removing latent failures from this safety subsystem.

cutoff period (1/freq) ->
75.4 uS

acceptable for feedback of power state, de-noises



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Redundant Monitored Power Switch/

File: PowerSwitch.kicad_sch

Title: InverterGen5 – Gate Driver Board

Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 8/8